Assessing the Spaceborne 183.31-GHz Radiometric Channel Geolocation Using High-Altitude Lakes, Ice Shelves, and SAR Imagery

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***Abstract*— The goal of this work is to perform the geolocation error assessment of the channel imagery at 183.31 GHz of the Special Sensor Microwave Imager/Sounder (SSMIS). The frequency around 183.31 GHz still represents the highest channel frequency of current spaceborne microwave and millimeter-wave radiometers. The latter will be extended to frequencies up to 664 GHz, as in the case of EUMETSAT Ice Cloud Imager (ICI). This use of submillimeter observations unfortunately prevents a straightforward geolocation error assessment using landmark-based techniques. We used SSMIS data at 183.31 GHz as a sub-millimeter proxy to identify the most suitable targets for geolocation error validation in very dry atmospheric conditions, as suggested by radiative transfer modeling. Using a yearly SSMIS data set, three candidates' landmark targets are selected: 1) high-altitude lakes and high-latitude bays using a coastline reference database and 2) Antarctic ice shelves using coastlines derived from Sentinel-1 Synthetic Aperture Radar (SAR) imagery. Data processing is carried out by using spatial cross correlation methods in the spatial frequency domain and performing a numerical sensitivity analysis to contour displacement. Cloud masking, based on a fuzzy-logic approach, is applied to automatically selected clear-air days. The results show that the average geolocation error is about 6.2 km for mountainous lakes and sea bays and 5.4 km for ice shelves, with a standard deviation of about 2.7 and 2.0 km, respectively. The results are in line with SSMIS previous estimates, whereas annual clear-air days are about 10% for mountainous lakes and sea bays and 18% for ice shelves.**

***Index Terms*— Image geolocation error, synthetic aperture radar (SAR) imagery, spaceborne millimeter-wave radiometers, surface landmarks.**

I. Introduction

THE next generation of submillimeter (sub-mm) wave radiometers will encompass the EPS-SG Ice Cloud Imager (ICI) mission, a sub-mm wave conical imager that is part of the EUMETSAT Polar System—Second Generation (EPS-SG) [1], [2] system. ICI will be on board of the Metop-SG satellite B series, to span a total operational lifetime over 21 years. ICI will represent a very important source of data for weather and climate applications. Its channels from 183.31 to 664 GHz will provide observations of ice clouds, supporting the validation and improvement of ice microphysics models adopted within numerical weather prediction (NWP) and climate models [3]. The use of sub-mm frequencies for observation of cloud ice properties has been investigated for more than two decades [4]–[8]. In this respect, being the first operational radiometer with submillimeter channels, ICI is opening not only new application domains but also new technological and technical issues.

Much experience has been accumulated so far on the geolocation error validation for satellite-based microwave radiometers at lower microwave frequencies (10–50 GHz, outside the absorption bands) by exploiting their strong difference in terms of surface emissivity between land and ocean. Global-scale coastlines are used as surface landmarks with a significant contrast in terms of the measured brightness temperature (BT). Comparing the latter with a reference coastline database [9], it is possible to assess the spaceborne sensor geolocation error. Purdy *et al.* [9] compared the shoreline obtained from WindSat satellite imagery with the World Vector Shoreline Data Bank II (WVS II). The position of the coastline is obtained taking the peak of the first derivative of radiometric data along scan and cross scan direction, after a cubic spline interpolation to obtain a more smoothed curve. Poe *et al.* [10] applied a similar method on Special Sensor Microwave Imager/Sounder (SSMIS) using data provided by spacecraft F-16. To assess the geolocation error of AMSR-E at 89 GHz, Wiebe *et al.* [11] and Heygster *et al.* [12] exploited the fact that when geolocation errors are present, the projected footprints have different shifts considering ascending or descending swaths. These differences are high along coastlines, optimizing the geolocation by retrieving the offset parameters that minimize the absolute values of the differences.

Berg *et al.* [13] used the BT difference between ascending and descending swaths to obtain the attitude error for SSM/I spacecraft. Bellerby *et al.* [14] also had a coastal-zone technique for attributing BT from the SSMI. Finally, Moradi *et al.* [15] corrected the pitch, yaw, and roll angles for Advanced Microwave Sounding Unit (AMSU) and Microwave Humidity Sounders (MHSs) minimizing the differences in BT

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between ascending and descending swaths. Along coastlines, the measured signal consists of radiation received from both land and water surfaces and Bennartz [16] proposed to use a high-resolution land–sea mask to infer the fraction of water surface for each measurement. He has developed a method to validate the geolocation accuracy using the convolution of land–sea masks that allows for channels that are sensitive to land/sea contrast. Han *et al.* [17] adapted this so-called "Land/sea Fraction Method" for the NOAA 16-18 satellites and also for ATMS on SNPP.

At frequencies beyond 150 GHz, the gaseous absorption can be so strong that the atmospheric transmittance, even in clear air, tends to be nearly zero. Moreover at sub-mm frequencies, the simulated emissivity of the surface water increases with frequency reaching values up to 0.65 around 400 GHz for horizontal polarization, surface winds around 20 m/s, and 50° incidence angle [18]. Therefore, the BT contrast between surface water and land bodies significantly diminishes, thus preventing conventional geolocation validation approaches based on surface landmarks. A way to circumvent these issues is to select the areas (and periods) with very low atmospheric water vapor concentration. Such surface landmarks, which should also exhibit a relatively large edge to be useful for geolocation assessment, are quite rare on Earth planet.

Within the open issues above mentioned, the goals of this work are twofold: 1) to exploit the current spaceborne radiometric channels around 183.31 GHz to experimentally look for water coastline and high-altitude surface targets that can be used for geolocation purposes and 2) to propose the landmark reference contours extracted not only from cartographic databases but also from high-spaceborne-resolution synthetic aperture radar (SAR) imagery. At 183-GHz frequency, useful surface landmarks can be found in very dry atmospheric conditions such as the polar Arctic and Antarctic areas in their winter time. In particular, the Antarctic region can provide relatively long edges less affected by the sea-ice formation. Polar regions are also very frequently sampled by Sun-synchronous satellites, thus offering a larger set of observations with respect to other regions, a feature quite critical for SAR missions. In this work, spaceborne radiometric data at 183.31 $\pm$ 6.6 GHz are obtained from the conically scanning microwave radiometer SSMIS in sun-synchronous, polar orbit.

This article is organized as follows. Section II describes the characteristics of selected landmark targets and satellite data set, considered here, and the best surface targets proposed for geolocation assessment. Section III introduces the processing steps for the geolocation validation and the sensitivity analysis. Section IV describes the application of the methodology for assessing the geolocation error for SSMIS using the considered surface landmarks. In Section V, conclusions are given.

II. Selected Landmarks and Satellite Data

Landmark targets should be characterized by high BT contrast to extract a contour, e.g., coastline, which is compared with a reference, e.g., shoreline database, to assess the geolocation accuracy. To find new topologies of landmark targets, we should focus on the main parameters that influence satellite BT (T_{B}). In the following, we assume that the antenna pattern deconvolution is already accomplished so that the antenna noise temperature T_{A} of the spaceborne radiometer is transformed into the satellite upward BT, named T_{Bsat}, incident upon the antenna itself.

In order to briefly clarify the landmark geolocation approach, for simplicity, let us consider a homogeneous isothermal atmospheric layer of thickness H with a small atmospheric albedo (thus neglecting the multiple scattering contribution) in the presence of a specular surface. Under these simplifying assumptions, it is possible to derive the analytical approximate solution of the integrodifferential radiative transfer equation for the total apparent BT incident upon the satellite antenna along the off-nadir viewing angles (θ, φ) as follows:

$$T_{\mathrm{Bsat}}(\theta, \varphi) = e_{\mathrm{s}} T_{\mathrm{s}} t + T_{\mathrm{Bup}} + T_{\mathrm{Bdw}}(1-e_{\mathrm{s}})t \tag{1}$$

with

$$T_{\mathrm{Bup}} = T_{\mathrm{Bdw}} = (1-\omega)T_0(1-t) \tag{1a}$$

$$t = e^{-\tau} = e^{-k_{\mathrm{e}}L} = e^{-k_{\mathrm{e}}H/\cos\theta} = e^{-(k_{\mathrm{a}}+k_{\mathrm{s}})H/\cos\theta} \tag{1b}$$

$$\omega = k_{\mathrm{s}}/k_{\mathrm{e}} \tag{1c}$$

where T_{Bup} is the atmospheric upwelling T_{B} (in this special case, equal to the downwelling one, T_{Bdw}), e_{s} is the surface emissivity, T_{s} the surface temperature, T_0 the constant atmospheric temperature, τ is the optical thickness, k_{e} the atmospheric extinction coefficient with k_{a} and k_{s} the absorption and scattering coefficients, respectively, ω the atmospheric albedo, L the atmospheric slant path (with $H = L\cos\theta$, with the θ nadir angle), and $e^{-(k_{\mathrm{e}}L)} = t$ the slant-path atmospheric transmittance. In the following, we can also suppose to deal with $T_{\mathrm{Bsat}}(\theta, \varphi)$, assuming that a deconvolution of the antenna pattern (and its side lobes) has been carried out at Level 1 data processing. Note that if the atmospheric layer is not isothermal, (1) can be easily generalized by: 1) substituting the constant atmospheric temperature with the mean radiative temperature T_{mr} and 2) the constant optical thickness τ with an integral of the extinction coefficient k_{e}. In this case, T_{Bup} will be generally less than T_{Bdw}.

Considering two contiguous pixels (x_i, y_i) and (x_j, y_j) of the acquired radiometric image (with $j = i \pm 1$), the BT spatial contrast ΔT_{Bsat} is given by

$$\Delta T_{\mathrm{Bsat}} = T_{\mathrm{Bsat}}(x_i, y_i) - T_{\mathrm{Bsat}}\left(x_j, y_j\right). \tag{2}$$

From (1), assuming a similar atmospheric layer with the same atmospheric transmittance t but with different surface emissivity e_{s} and temperature T_{s} of the two considered pixels, we can rewrite (2) as follows:

$$\Delta T_{\mathrm{Bsat}} = t[e_{\mathrm{s}}(x_i, y_i)T_{\mathrm{s}}(x_i, y_i) - e_{\mathrm{s}}(x_j, y_j)T_{\mathrm{s}}(x_j, y_j)]. \tag{3}$$

Equation (3) shows that ΔT_{Bsat}, due to two adjacent pixels, essentially depends on the surface emissivity and surface temperature weighted by atmospheric transmittance t.

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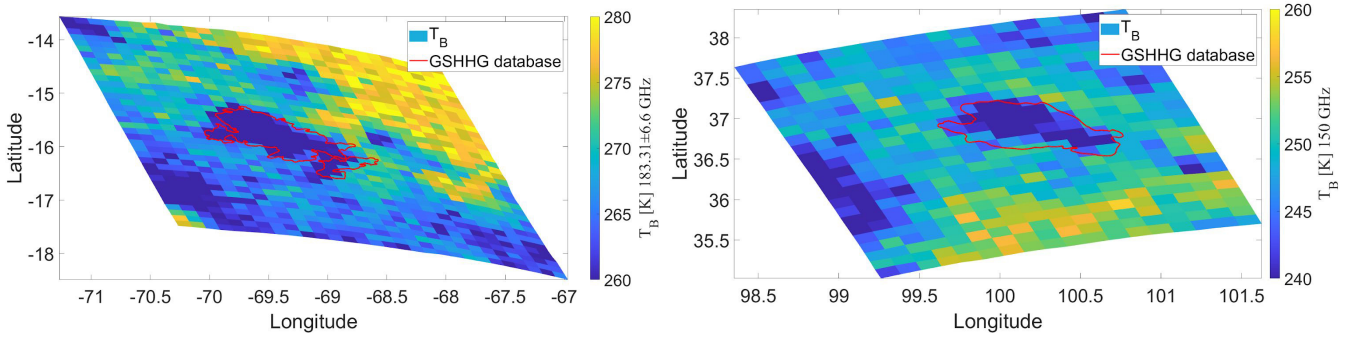


Fig. 1. (Left) Brightness temperature image at 183.31 ± 6.6-GHz H-pol over the Lake Titicaca with SSMIS F17 on May 31, 2016. (Right) Brightness temperature image at 183.31 ± 6.6-GHz H-pol over the Qinghai Lake with SSMIS F17 on December 2, 2016. The red line represents the boundary between lake and land, provided by level 2 of the GSHHG shoreline database [18].

If the atmosphere is too opaque, its transmittance t can be much less than 1, thus reducing the BT spatial contrast and impeding to extract a clean contour from the surface landmark. This consideration is prompting us to look for Earth regions where the average atmospheric transmittance and surface emissivity spatial contrast are as high as possible. High-altitude lakes, high-latitude bays, and Antarctic ice shelves are the best candidates for this goal. All these targets in clear-air conditions and drier seasons typically exhibit a ΔT_{Bsat} contrast larger than 10 K.

A. High-Altitude Lakes

At 183.31 ± 6.6 GHz, low optical thickness, particularly in regions of high elevation, increases the surface contribution as observed by the spaceborne radiometer, making high-altitude features suitable targets. Taking into account (3), we focused on areas in which adjacent pixels have different surface emissivity and specifically on high-altitude lakes that have water-land pixels. In this work, we focused specifically on two targets for geolocation error assessment, Lake Titicaca for southern hemisphere and Qinghai Lake for northern hemisphere.

The Lake Titicaca is a large, deep lake in the Andes on the border of Bolivia and Peru with an elevation of 3812 m. This lake has a maximum length of about 204 km and a width of about of 65 km with a total surface of about 8372 km^2. Compared to the SSMIS pixel at 183.31 ± 6.6 GHz (about 13 km), it is possible to successfully detect it and extract its contour.

The left of Fig. 1 shows the BT image at 183.31 ± 6.6 GHz at H polarization over the lake from SSMIS F17 on May 31, 2016. The lake shape, clearly indicated by the coastline contour [19], appears from the contrast between the water emissivity (about 0.55 on average, from [18]) and land emissivity (about 0.90 on average, from [20]). The lake appears colder with respect to the surrounding land with a contrast of about 12 K in dry conditions. Areas with higher T_{B} around 280 K in the north-east of the lake are due to the vegetation cover at lower altitudes within the Andes mountains.

The Qinghai Lake is a large lake in China with a surface of about 4500–6000 km^2 and an altitude of about 3200 m. Its maximum length is about 105 km and the width is about 65 km. It is smaller than Titicaca but large enough to extract its contour. The right of Fig. 1 shows the BT image at 183.31 ± 6.6 GHz at H polarization over the Qinghai Lake with SSMIS F17 on December 2, 2016. The Qinghai Lake contour is also highlighted by the shoreline database line. The lake appears colder than the surrounding land due to water lower emissivity at 183.31 GHz, with a contrast of about 13 K on average in dry conditions.

B. High-Latitude Sea Bays

Taking advantage of the difference between land and water emissivities, we can also explore surface targets such as high-latitude bays appearing as internal water bodies having a contour as close as possible.

One of these targets is the Hudson Bay in the northern hemisphere. It is a large body of saltwater in northeastern Canada with a surface area of 1 230 000 km^2. Fig. 2 shows the south-western area of the Hudson Bay, where the ΔT_{Bsat} contrast between sea and land is about 10 K on average in dry conditions.

During colder seasons, the land region around the Hudson Bay is typically covered by snow from the end of October to May [21], whereas sea ice is formed inside the bay. In these conditions, the snow cover shows a lower surface emissivity (about 0.75 on average, from [20]) with respect to the first year sea ice emissivity (about 0.85 on average, from [20] and [22]) at 183.31 ± 6.6 GHz at H polarization. Following (3), in these conditions, the Hudson Bay coastline is a good candidate to provide a BT contrast sufficient to identify the shoreline from SSMIS radiometric images.

C. Antarctic Ice Shelves

Promising targets are also identified in the ice cover shorelines. More specifically, in this work, we propose to use ice shelves in the Antarctic regions to support evaluating the geolocation error accuracy. These shelves are thick suspended platforms of ice with enough BT contrast to extract the contour of the ice shoreline. The ice covers the ground (grounded ice) and it extends into the ocean, with the lower part that detaches

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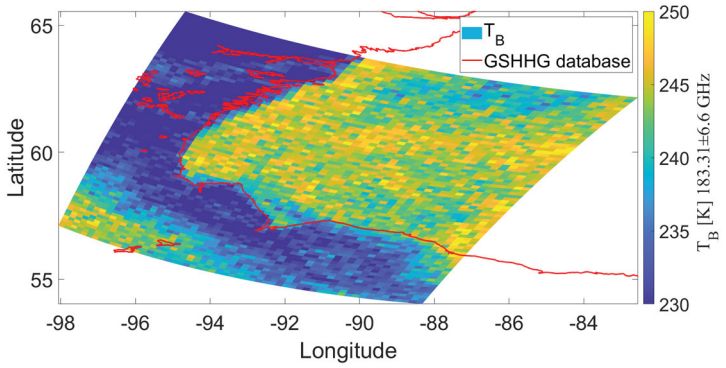


Fig. 2. Brightness temperature image at 183.31 ± 6.6-GHz H-pol over the Hudson Bay with SSMIS F17 on March 4, 2016. The red line represents the boundary between lake and land, provided by level 2 of the GSHHG shoreline database [18].

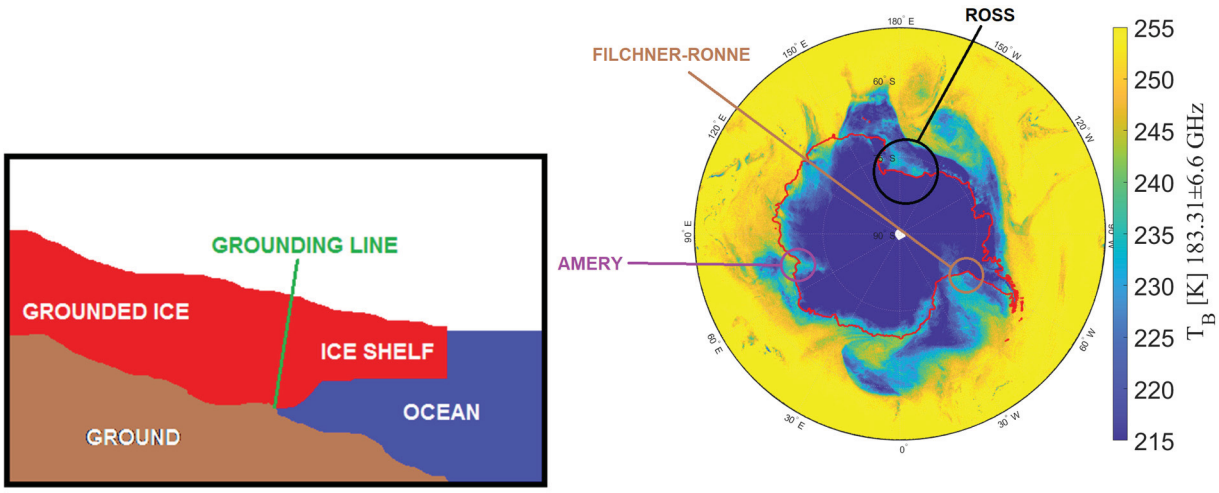


Fig. 3. (Left) Generic scheme of Antarctica ice shelves coastline. Ice is represented in red, and the ice shelf is the floating platform not attached to the ground below. (Right) Brightness temperature over Antarctica region from SSMIS F17 at 183.31 ± 6.6 GHz on August 2, 2016 (horizontally polarized) using ten swaths. The brown circle highlights the Filchner–Ronne ice shelf, the black circle indicates the Ross ice shelf, and the violet circle indicates the Amery ice shelf. The red line represents the boundary between Antarctica ice and ocean, provided by level 5 of the GSHHG shoreline database [18].

from the ground (grounding line), creating a suspended platform of ice (ice shelf), as shown in Fig. 3 (left). Fig. 3 also shows the BT over Antarctica observed from SSMIS F17 at 183.31 ± 6.6 GHz. The swath contains two large shelves, the Ross and Filchner–Ronne ice shelves. Both shelves appear colder (lower than about 225 K) with respect to the adjacent sea (about 240 K).

Fig. 4 details the BT over the three shelves from SSMIS F17 at 183.31 ± 6.6 GHz (H). The three images provided by SSMIS confirm that ice shelves are colder with respect to sea and this BT contrast allows to obtain a contour line from radiometric imagery.

However, the ice coastline can change its shape along seasons or it is also possible that pieces of shoreline can collapse creating icebergs, with a consequent change in shape. Considering the biggest ice shelf, the Ross, as reported in [23], the ice velocity ranges from a few meters per year to several hundred meters per year in ice streams. Ice velocity increases as the ice moves seaward, reaching 1 km per year in the central portions of the ice front. As such, it is necessary to account for this aspect or neglect it focusing on the shoreline with lower variability, e.g., lateral parts of the shelves.

D. Satellite Sensors and Data

The main characteristics of the data set used in this work are listed in Table I. Data consist in four main sources.

1) SSMIS image data from NOAA Climate Data Record (CDR) developed at Colorado State University (CSU), providing the T_B images around at 183.31 ± 6.6 GHz (H) [24] for the whole of 2016.
2) SAR image data from the European Space Agency (ESA) Sentinel 1A mission, available in Extra-Wide Swath Mode with a 400-km swath at 20 m × 40 m spatial resolution. SAR imagery is characterized by speckle noise and adjacent pixels can have high contrast. To reduce the speckle, we use the Level 1 Ground Range Detected (GRD) SAR products with medium resolution (MR). GRD data consist of focused SAR data

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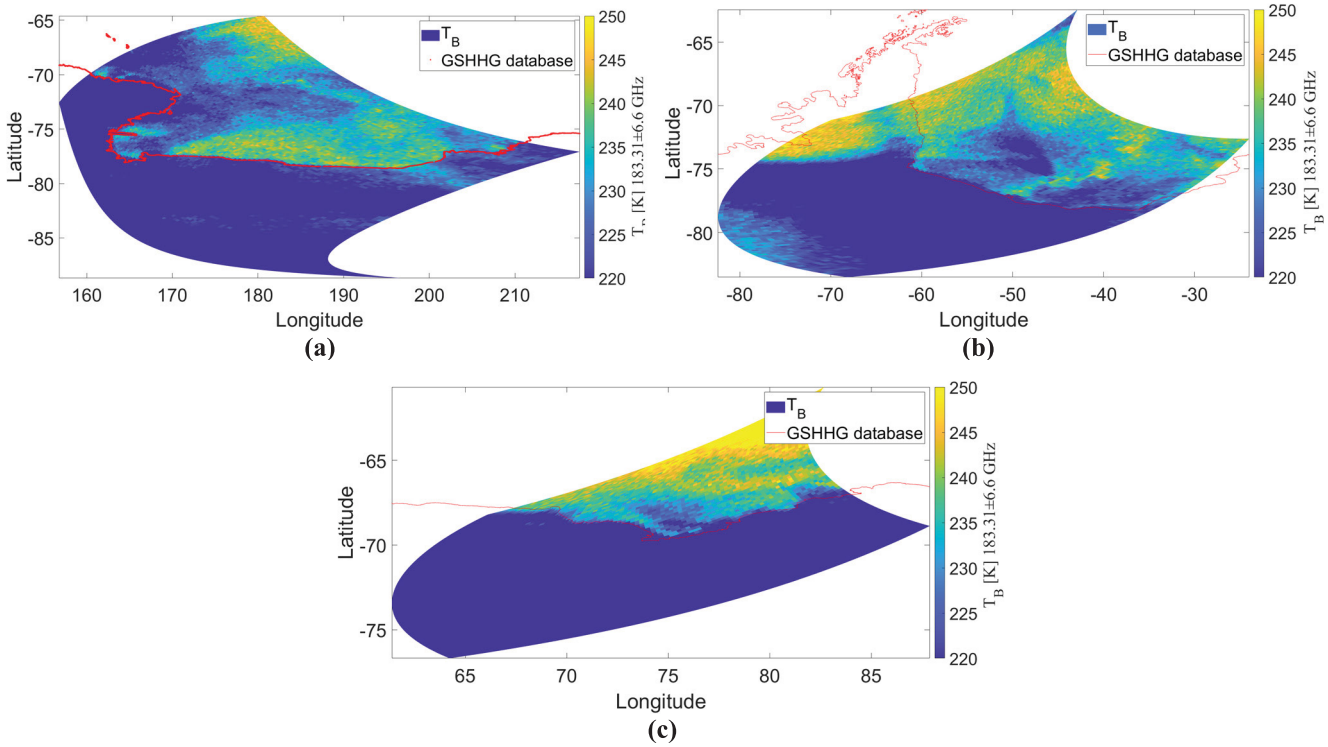


Fig. 4. (a). Brightness temperature image at 183.31 ± 6.6-GHz H over the Ronne ice shelf with SSMIS F17 on June 7, 2016. (b) Brightness temperature image at 183.31 ± 6.6 GHz H over the Filchner–Ronne ice shelf with SSMIS F17 on May 20, 2016. (c) Brightness temperature image at 183.31 ± 6.6 GHz H over the Amery ice shelf with SSMIS F17 on June 17, 2016.

TABLE I
SUMMARY OF THE USED DATA

Reference source	Characteristics	Spatial resolution	Pre-processing needed
SSMIS	Brightness Temperature (183.31±6.6 GHz)	13 x 16 km	no
SAR	Level-1 GRD	10-40 m	yes
GSHHG	GSHHG with full resolution	40 m	no
DEM	GTOPO 30	30 arc seconds (~1 km)	no

that have been detected, multilooked, and projected to ground range using an Earth ellipsoid model.

3) Coastline contour data from the Global Self-consistent, Hierarchical, High-resolution Geography (GSHHG) database in full resolution [19].
4) The GTOPO30 is a digital elevation model (DEM) with a resolution of 30 arcsec (about 1 km) [25].

III. Geolocation Assessment Methodology

The general scheme of the adopted methodology is described in Fig. 5, whose modules will be illustrated in Section III-A. Sensitivity analysis is discussed in Section III-B.

A. Target Contour Matching

The target contour matching (TCM) [26] is the proposed methodology to validate the geolocation at 183.31 ± 6.6 GHz channel and its major steps are described below.

1) Input-Screened Data: Spaceborne MW imagery, containing a landmark target, represents the starting point of the TCM methodology. Within a relatively extended period, e.g., one year, there are several available satellite overpasses, depending on the latitude of the target and the instrument swath. As previously discussed, not all landmark overpasses can be used to assess the geolocation. These can be included in the analysis fulfilling the following conditions:

1) clear sky or limited cloudy conditions in the selected area;
2) very dry conditions with little water vapor; and
3) enough BT contrast of the target to extract a contour.

As a screening criterion to select only useful overpasses of the target landmark, we propose a relatively straightforward fuzzy-logic approach, which is described in the Appendix. Following the left branch in Fig. 5, after the defuzzification step, a further process is the geolocation adjustment of spaceborne radiometric data in case of high-altitude targets.

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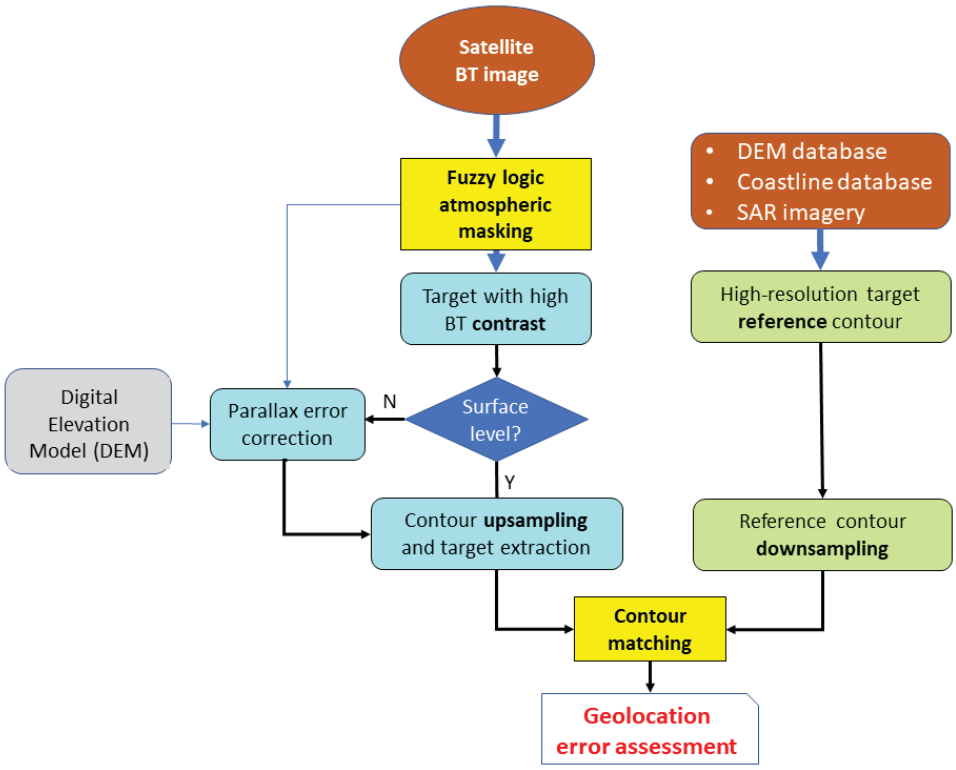


Fig. 5. Logical scheme of the proposed methodology to validate the geolocation accuracy using landmark targets (target-contour matching algorithm).

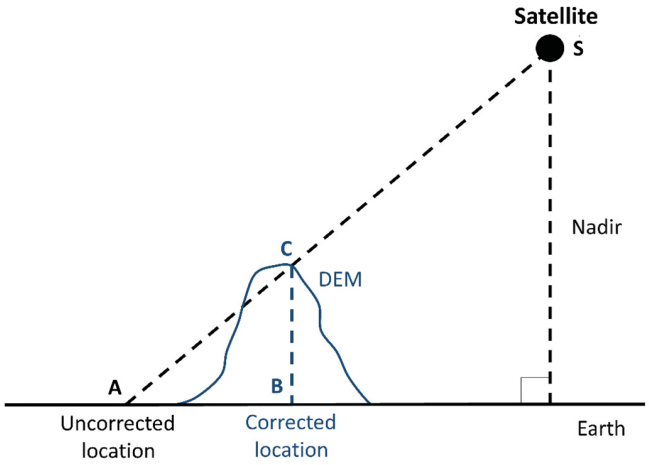


Fig. 6. Parallax error correction using DEM. Point S represents the satellite position, and A indicates the uncorrected coordinates on ellipsoid. Point C is the intersection between orography and satellite line of sight. Point B indicates the corrected coordinates after parallax error correction.

This step is necessary for mountainous areas since geolocation of SSMIS is performed above the terrestrial ellipsoid. Conversely, it is not performed when the target is at sea level (as in the case of ice shelves and bays).

2) Parallax Error Correction: In the presence of mountains, the line of sight intersects the orography of Earth region (point C) before intersecting the ellipsoid (point A). Fig. 6 shows a schematic of the parallax error correction, applied to each pixel not at the sea level. A correction is needed to obtain the true latitude and longitude coordinates of the pixel, matching point B in the same figure. In this work, the GTOPO30 DEM is used to extract the intersection between the satellite line-of-sight and the orography. After this step, the whole image is shifted, depending on ascending or descending orbit and the position of the landmark target in the satellite swath.

To reduce the distortion in the polar regions, BT image has been further reprojected on a polar stereographic map in the x–y coordinates. Both coordinates are obtained from latitude (φ) and latitude (λ) according to the following equations [27]:

$$x = \rho \sin(\lambda - \lambda_0) \tag{4a}$$

$$y = -\rho \cos(\lambda - \lambda_0) \tag{4b}$$

where

$$\rho = a m_c t / t_c \tag{5a}$$

$$\lambda_0 = 70^\circ \tag{5b}$$

$$t = \tan(\pi/4 - \varphi/2) / \left[\frac{1 - e\sin(\varphi)}{1 + e\sin(\varphi)}\right]^{e/2} \tag{5c}$$

$$m_c = \cos(\varphi_c) / \sqrt{(1 - e^2(\sin(\varphi_c))^2)} \tag{5d}$$

where $\varphi_c = 0^\circ$ and t_c are obtained from (5c) by imposing $\varphi = \varphi_c$. This step is not applied in the geographic areas far from the poles, where the latitude–longitude grid is sufficiently accurate.

3) Spatial Scaling: Since the satellite microwave radiometric images have a lower spatial resolution compared to the

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expected subpixel geolocation accuracy, a spatial downscaling is applied. A finer resolution of the BT image is reached by using a factor of about 3 on an evenly spaced grid in lat/lon (or x, y) and adopting a spatial cubic interpolation. The output pixel resolution of each BT image at 183.31 GHz is then about 4.3×5.3 km^2.

On the other hand, the reference contours are typically available at much higher spatial resolution (e.g., between 10 and 50 m) so that an upscaling to the BT downscaled spatial resolution is needed. In case of high-altitude lakes and high-latitude bays, the upscaling procedure is applied to the GSHHG shoreline, as already mentioned.

For ice shelves, we propose to use SAR images as a reference in order to monitor the variations of ice shelves shoreline. SAR data, described in Section II-A, can exhibit a reduced speckle and in this work, at polar region, the preferable acquisition mode is Extra Wide (EW) Swath with a spatial resolution of 40 m $\times$ 40 m (in case of MR). Nevertheless, resolution is still high compared to the radiometer geolocation accuracy requirements so that the error associated with SAR image data is negligible. SAR data have been processed using Sentinel Application Platform (SNAP), provided by ESA [28]. The workflow consists in the following steps: 1) apply orbit file: to provide more accurate satellite position and velocity information; 2) thermal noise removal: to reduce the noise effects in the intersubswath texture; 3) border noise removal: to compensate the change of Earth's curvature; 4) calibration: the pixel values can be related to the radar backscatter of the scene (σ^0); and 5) multilook algorithm: to further reduce the speckle, it is possible to average adjacent pixels, but the final spatial resolution further decreases. The final step consists of extracting the reference ice-shelf contour from the SAR processed image. To this aim, we have used the Canny approach for edge detection, briefly described in the next paragraph.

4) Contour Extraction: The next TCM step is the extraction of a contour from the SMMIS-screened image. The Canny technique can be efficiently applied to obtain a contour line [28]. This method mainly consists in the following steps:

1) convolution with Gaussian Filter to remove the noise, obtaining a smoothed image;
2) find the intensity gradients of the image;
3) apply nonmaximum suppression to get rid of spurious response to edge detection;
4) apply double threshold to determine the potential edges; and
5) hysteresis thresholding, obtaining the detection of edges.

5) Contour Matching: The BT-extracted contour is then cross-correlated with the reference contour, derived either from GSHHG shoreline or from SAR imagery, in order to estimate the geolocation error accuracy. Note that, to correlate the two extracted curves, it is necessary to have them on the same grid. Thus, in the case of SAR-derived contour, the latter is projected on the same polar stereographic map used for SSMIS radiometric image, using the nearest neighbor approach.

Finally, in order to correlate the two extracted curves, obtaining directly a displacement with subpixel accuracy, we can use an efficient subpixel image registration by spatial cross correlation [30]. This approach demonstrates to have shorter computation times and significantly reduced random access memory requirements.

TCM algorithm needs different samples to assess the geolocation error for each target so that the proposed target should be visible several times during the period of interest. The detailed analysis about the requested number of samples for the chosen targets is given in Section IV.

B. TCM-Expected Spatial Accuracy

To evaluate the performance of the proposed TCM methodology, we can perform a numerical sensitivity analysis by imposing an arbitrary known geolocation shift along latitude and/or longitude and then evaluate the retrieved TCM geolocation error. For brevity, we show this test only for the Filchner–Ronne ice shelf and Qinghai Lake targets.

For each target and scene, using the original cloud-masked images of the data set, we have generated 441 scenes adopting an arbitrary geolocation error with a discrete step of 0.01° (the number 441 comes from the imposed 21 shifts along both latitude and longitude). For the Filchner–Ronne ice shelf tests, we have selected the cloud-masked images of two different days, i.e., June 21, 2016, and August 2, 2016, with high values of $I(\varepsilon, \Delta T_{\mathrm{Bm}})$ where I is the inference function described in detail in the Appendix, as a function of the geolocation error ε and mean BT contrast ΔT_{Bm}. In particular, $I(\varepsilon, \Delta T_{\mathrm{Bm}})$ is 0.86 for June 21, 2016, and $I(\varepsilon, \Delta T_{\mathrm{Bm}})$ is 0.96 for August 2, 2016. The geolocation error estimate has been then computed for all the generated shifts.

The results are provided in Fig. 7 for the Filchner–Ronne (top figures) and Qinghai Lake (bottom figures) as scatterplots between the imposed error and the retrieved one. The results for the Filchner–Ronne shelf on June 21, 2016, and August 2, 2016, show a mean difference between the imposed and retrieved errors of about 0.30 and 0.25 km, respectively, with a standard deviation of about 0.52 and 0.60 km. For the Qinghai Lake, a mean value of 0.23 km and a standard deviation of 0.86 were obtained for the November 3, 2016 ($I(\varepsilon, \Delta T_{\mathrm{Bm}}) = 0.66$) cloud-masked image and a mean value of 0.27 km and a standard deviation of 0.58 for the November 10, 2016 ($I(\varepsilon, \Delta T_{\mathrm{Bm}}) = 0.56$) cloud-masked image.

The previous numerical tests show that the proposed TCM methodology can reconstruct with a fairly good accuracy the imposed geolocation errors, with a standard deviation lower than 1 km independently of the target. Indeed, this estimated standard deviation is lower than the requirement for an adequate error geolocation assessment of a spaceborne MW radiometer.

IV. Results

In this section, the results will be presented separately for both typologies of surface targets.

A. Geolocation Error Assessment for Lakes and Bays

The entire data set of SSMIS F17 during 2016 consisted of 5108 full orbits, which provided a total of 7550 overpasses subdivided among the different targets. As discussed

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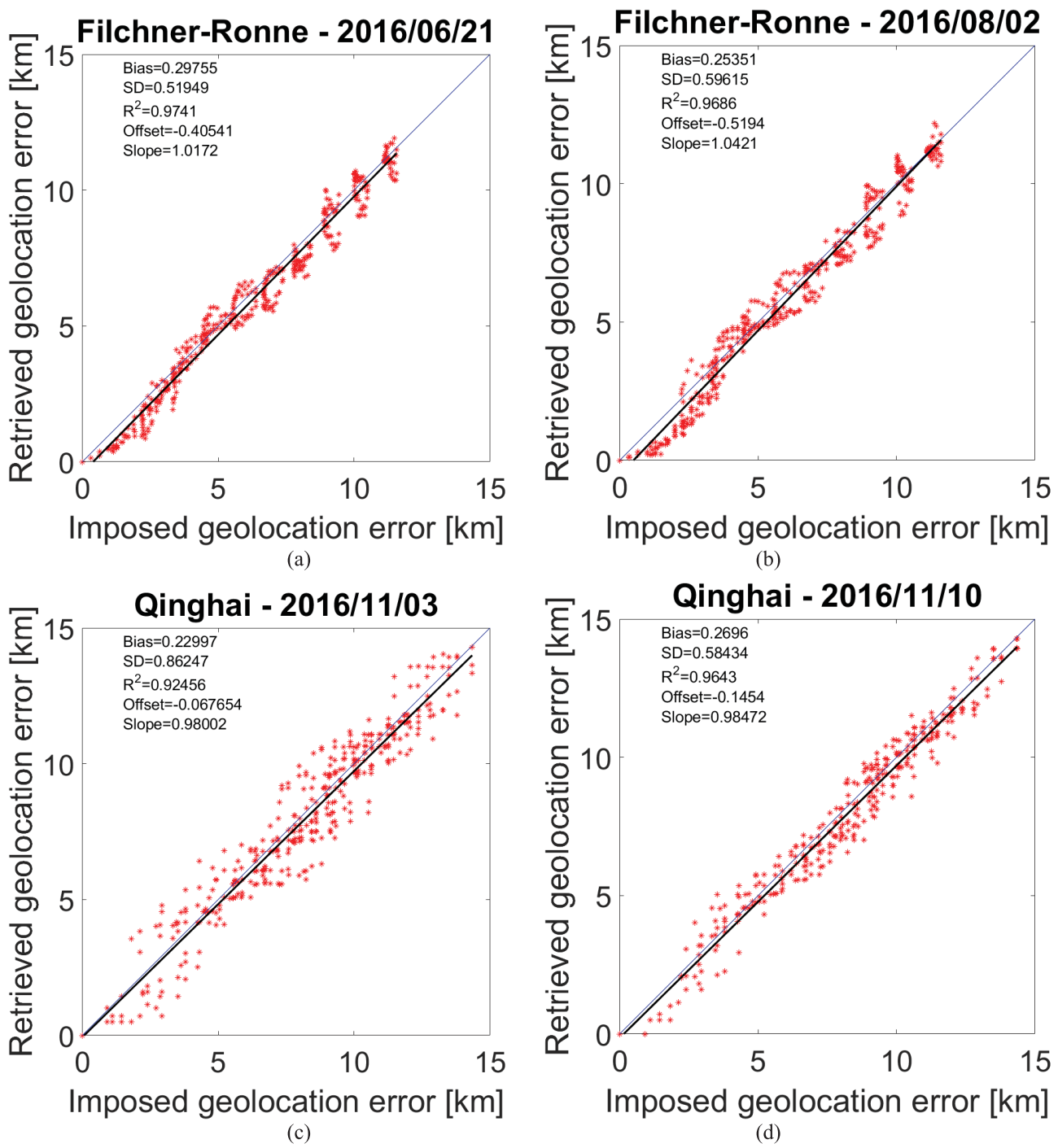


Fig. 7. Sensitivity analysis of geolocation error assessment procedure based on TCM. Scatterplot between imposed geolocation error and retrieved one for Filchner–Ronne ice shelf target on (a) April 15, 2016 and (b) May 20, 2016. Scatterplot between imposed and retrieved geolocation error using the Qinghai Lake as target on (c) November 10, 2016, and (d) November 25, 2016.

in Section III-A, however, only a limited number of these can be effectively used for the geolocation assessment. After the cloud-masking defuzzification step, a total number of 41 cases for the Lake Titicaca and 84 for the Qinghai Lake were selected during 2016 from SSMIS 17. The same defuzzification step applied to the Hudson Bay provides 62 useful samples for 2016.

The number of selected cases for these three targets is grouped in Table II according to the month. The two lakes are visible in complementary annual periods. The Qinghai Lake in the northern hemisphere is visible only in colder dry months from October to May, while the Lake Titicaca is visible on clear-sky dry conditions from April to November. The Hudson Bay is in the northern hemisphere so that it is visible in the

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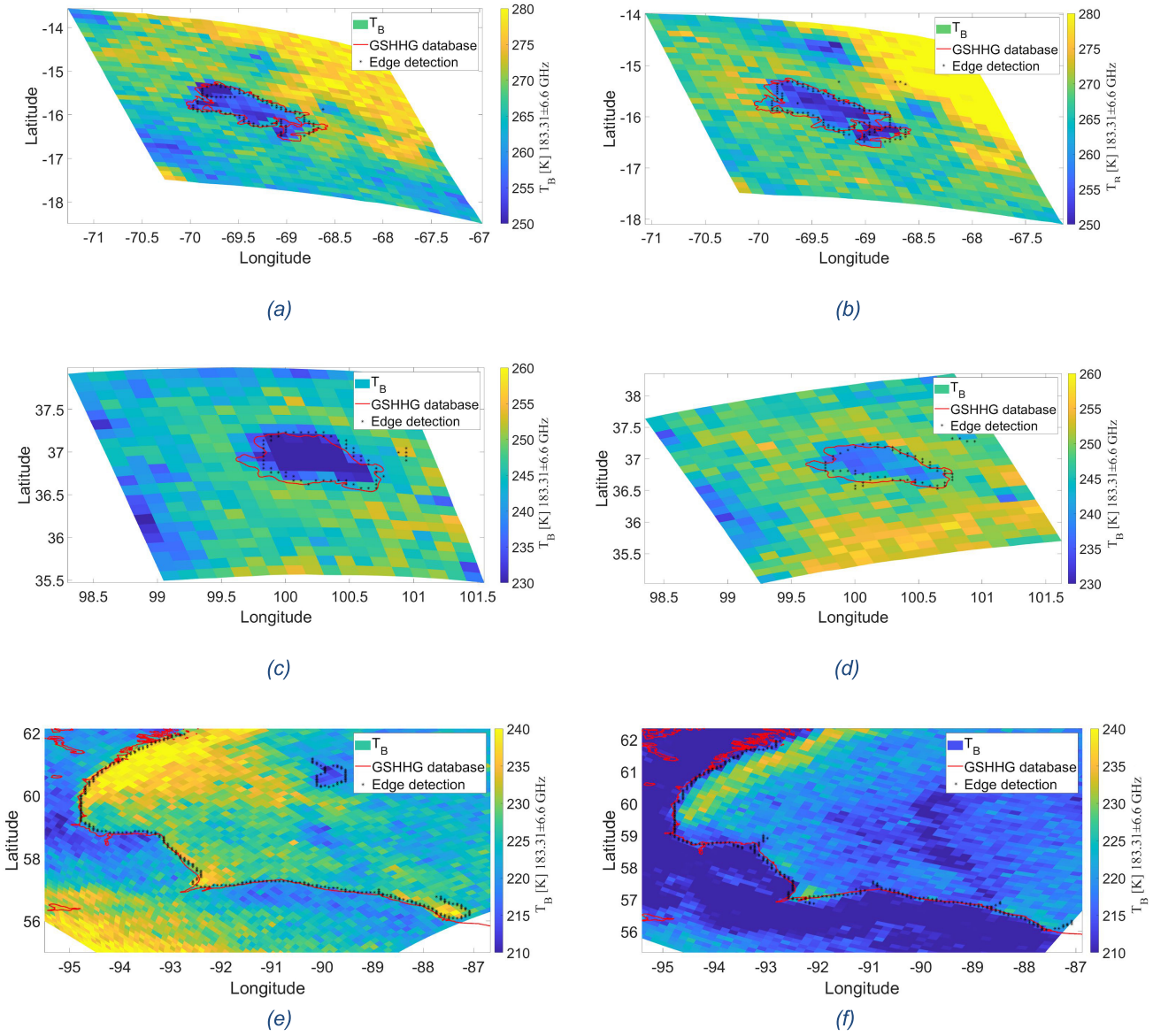


Fig. 8. Brightness temperature image at 183.31 ± 6.6-GHz H over the Lake Titicaca with SSMIS F17 on (a) May 31, 2016, and (b) July 31, 2016. Brightness temperature image at 183.31 ± 6.6-GHz H over the Qinghai Lake with SSMIS F17 on (c) December 1, 2016 and (d) December 2, 2016. Brightness temperature image at 183.31 ± 6.6-GHz H over the Hudson Bay with SSMIS F17 on (e) January 27, 2016, and (f) February 11, 2016. The red markers represent the GSHHG shoreline database and black markers are provided by applying the Canny edge detection on the radiometric images.

TABLE II

NUMBER OF VISIBILITY CASES DURING 2016 FOR HIGH-ALTITUDE LAKE APPLYING THE CLOUD-MASKING ALGORITHM

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Lake Titicaca	0	0	0	2	5	8	8	9	4	2	3	0	**41**
Qinghai Lake	5	1	5	13	5	0	0	0	0	6	22	27	**84**
Hudson Bay	13	17	10	3	0	0	0	0	0	0	0	19	**62**

same period of the Qinghai Lake. Examples of SSMIS BT over the Lake Titicaca, Qinghai Lake, and Hudson Bay are shown in Fig. 8. As discussed in Section II, the lakes are colder with respect to the land. The GSHHG shoreline database is highlighted by a red line. In Fig. 8, the contour (black dots), extracted by using the Canny algorithm, is not superimposed to the reference outline, thus indicating a geolocation displacement in the radiometric imagery.

By applying the TMC method to the cases of Table II, using the GSHHG as the reference contour as discussed in Section III-A, the results are summarized in Table III. The mean geolocation error for the Lake Titicaca is 6.53 km,

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TABLE III
RESULT FOR HIGH-ALTITUDE LAKE ALONG 2016

Geolocation assessment parameter	Value [km]		
	Titicaca Lake	Qinghai Lake	Hudson Bay
Geolocation accuracy average [km]	6.53	5.86	6.29
Geolocation accuracy standard deviation [km]	2.82	2.59	2.89

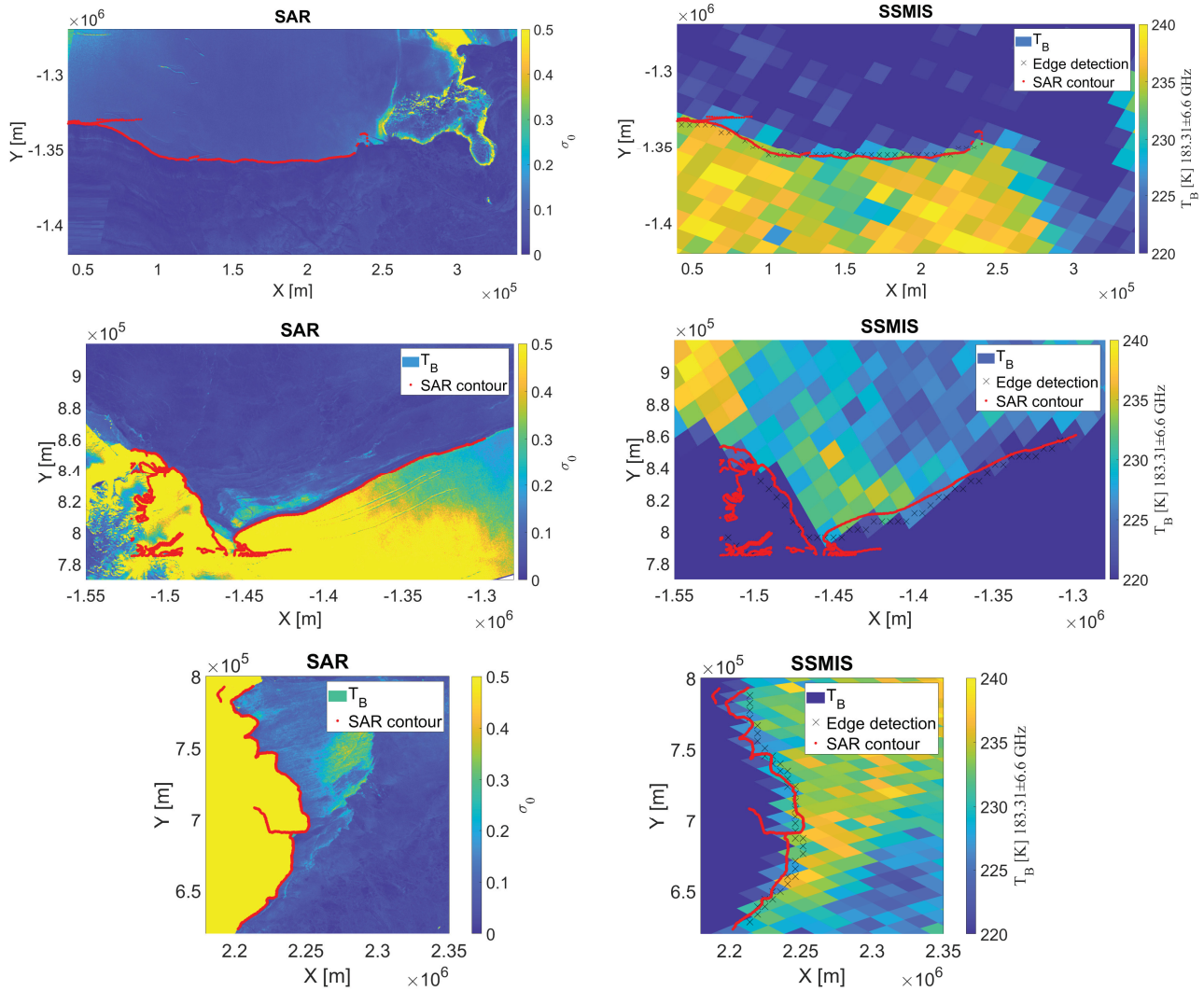


Fig. 9. (Left) Example of the SAR image used as reference within TCM. (Right) Reported example of brightness temperature at 183.31 ± 6.6-GHz H from SSMIS F17. (Top) Images for the Ross ice shelf. (Center) Images for the Filchner–Ronne ice shelf. (Bottom) Images for the Amery ice shelf.

for the Qinghai Lake is 5.86 km, and for the Hudson Bay is 6.29 km. The standard deviation values are similar and specifically 2.82 km for the Lake Titicaca, 2.59 for the Qinghai Lake, and 2.89 km for the Hudson Bay. These tests show that, using the GSHHG contours, we can obtain very similar results considering high-altitude lakes and high-latitude bays. In particular, the overall averaged value of the geolocation error is about 6.2 km with a standard deviation of about 2.8 km.

B. Geolocation Error Assessment for Ice Shelves

The Antarctic ice shelves, being in a polar area, are visible several times per day so that SSMIS data set is larger than midlatitude and equatorial ones. Indeed, the Ross ice shelf is yearly visible 599 times, the Filchner–Ronne one 387 times and the Amery one 153 times, as shown in Table IV. Note that these targets are more visible from April to October because this period coincides with the austral winter colder months

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TABLE IV
NUMBER OF VISIBILITY CASES DURING 2016 FOR ANTARCTIC ICE SHELVES LAKE APPLYING THE CLOUD-MASKING ALGORITHM

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Ross	0	0	15	67	57	109	114	78	81	61	17	0	**599**
Filchner-Ronne	0	0	14	62	65	46	78	65	42	15	0	0	**387**
Amery	5	5	20	22	13	32	19	11	15	7	4	0	**153**

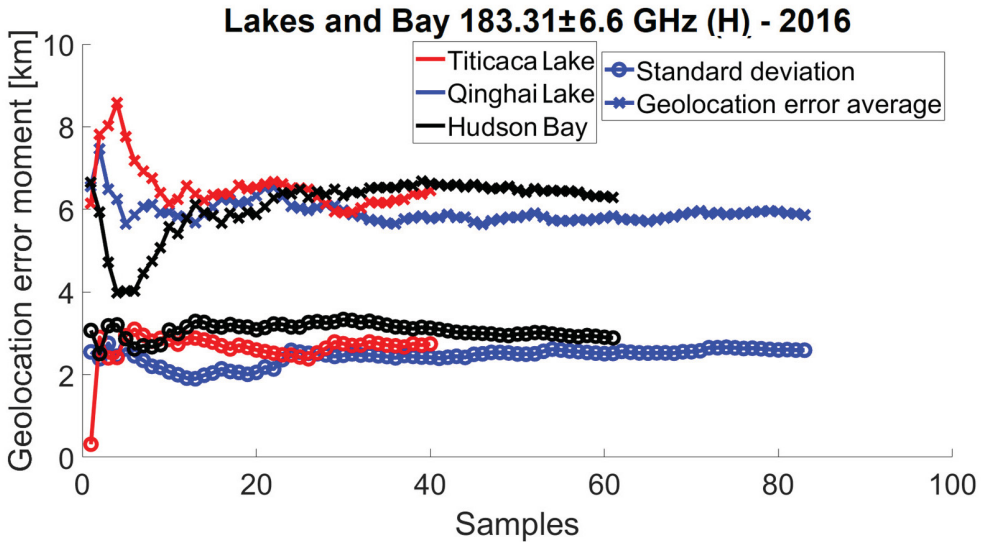


Fig. 10. Geolocation error average (x markers) and standard deviation (o markers) for the Qinghai Lake (red line), Lake Titicaca (blue line), and Hudson Bay (black line) using SSMIS F17 at 183.31 ± 6.6 GHz (H) along 2016.

with an integrated water vapor content much lower than the austral summer one.

As discussed in Section III-A, the SAR images have been used as reference. Fig. 9 shows the examples of TB images from SSMIS F17 over the Ross ice shelf, the Filchner–Ronne, and the Amery one. The corresponding SAR reference images are also shown. The SAR backscattering coefficient (σ^0) is, as expected, larger over the ice cap than over ocean due to the shelf surface roughness. This difference creates a sharp contrast along the ice shoreline allowing a clean contour extraction used for geolocation error assessment. The ice shoreline derived from SAR image and extracted contour from SSMIS data are also shown in the BT maps of Fig. 9.

The overall results, obtained by applying the TMC method to the defuzzified data set of year 2016, are summarized in Table V. A geolocation average error of about 4.17 km with about 2.16 km as the standard deviation is found for the Ross ice shelf, about 5.75 km average with about 2.35 km of the standard deviation for the Filchner–Ronne ice shelf, and a mean value of about 6.25 km with about 1.55 km as the standard deviation for the Amery ice shelf. It is worth noting that the geolocation error results are slightly different for the three shelves due to the natural shape of these targets. As a matter of fact, the lake contours represent a closed line so that they are adequate to detect the mean displacement along all directions. Conversely, the shelf contours are open curves, thus presenting the disadvantage to have preferable directions and well representing the displacement only in a direction perpendicular to its shape.

Considering the overall error mean value for all the Antarctic ice shelves, we obtain a geolocation error of about 5.4 km with a standard deviation of about 2.0 km. The standard deviation is lower in this case than for lakes and bays mainly because a larger data set for polar regions is available with respect to midlatitude and equatorial targets.

C. Impact of Target Sample Number

This section shows the sensitivity of the geolocation error assessment to the number of samples for all selected landmark targets. This is a critical issue especially for a calibration and validation activity, which should be performed in a limited period of time (e.g., during the satellite commissioning phase).

Fig. 10 reports the behavior of the average geolocation accuracy and relative standard deviation with respect to the number of available images for the three targets described in Section IV-A. In order to estimate the minimum number of samples necessary to have the convergence of the results, we can introduce the relative difference of the average mean

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TABLE V
RESULT FOR THE ANTARCTIC ICE SHELVES ALONG 2016

Geolocation assessment parameter	Value [km]		
	Ross	Filchner-Ronne	Amery
Geolocation accuracy average [km]	4.17	5.75	6.25
Geolocation accuracy standard deviation [km]	2.16	2.35	1.55

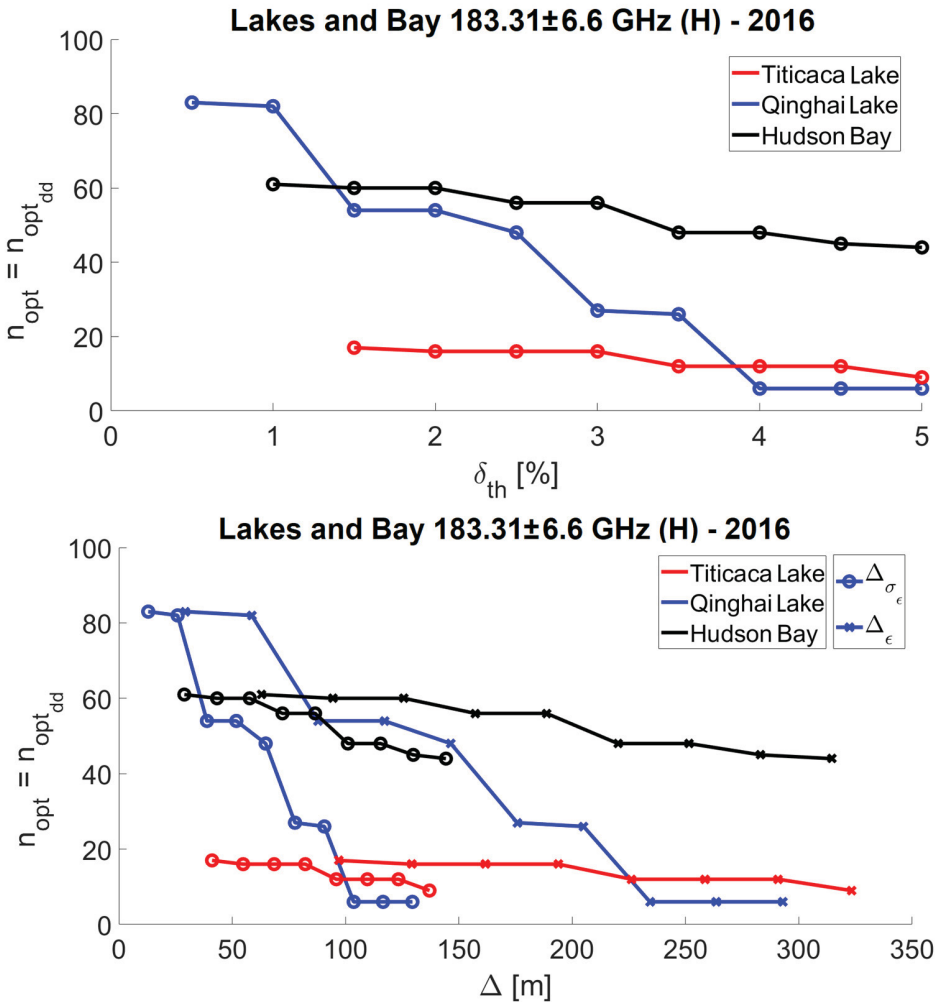


Fig. 11. (Top) Relation between the minimum number of necessary samples (n_{opt}) and δ_T threshold. (Bottom) Relation between the minimum number of necessary samples (n_{opt}) and Δ_ε with 'x' markers and between the minimum number of necessary samples (n_{opt}) and $\Delta_{\sigma_\varepsilon}$ with "o" markers. The results are shown for the Qinghai Lake with red line, for the Lake Titicaca with blue line, and for the Hudson Bay with black line.

error $\delta_\varepsilon(n)$ and the relative difference of the average standard deviation $\delta_\sigma(n)$ as

$$\delta_\varepsilon(n) = \frac{\varepsilon(n) - m_{\text{e}}}{m_{\text{e}}} 100 \tag{6a}$$

$$\delta_{\sigma_\varepsilon}(n) = \frac{\sigma_\varepsilon(n) - m_\sigma}{m_\sigma} 100 \tag{6b}$$

where $\varepsilon(n)$ is the average of the geolocation displacement obtained considering the first n samples, m_ε is the average of the geolocation displacement obtained considering all samples, $\sigma_\varepsilon(n)$ is the average of the standard deviation of geolocation displacement obtained considering the first n samples, and m_σ is the average of the standard deviation of geolocation displacement obtained considering all samples

Note that both $\delta_\varepsilon(n)$ and $\delta_\sigma(n)$ are percentage values. Setting arbitrary thresholds for both relative differences, we can impose the convergence of the results when it happens

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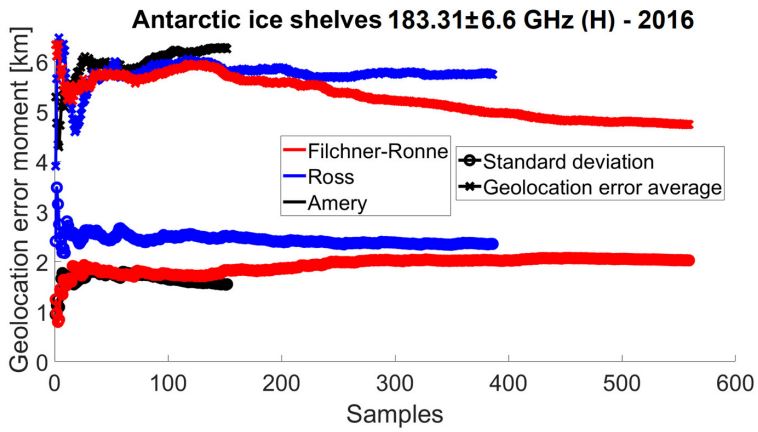


Fig. 12. Geolocation error average (x markers) and standard deviation (o markers) for the Ross ice shelf (red line), Filchner–Ronne ice shelf (blue line), and Amery ice shelf (black line) using SSMIS F17 at 183.31 ± 6.6 GHz (H).

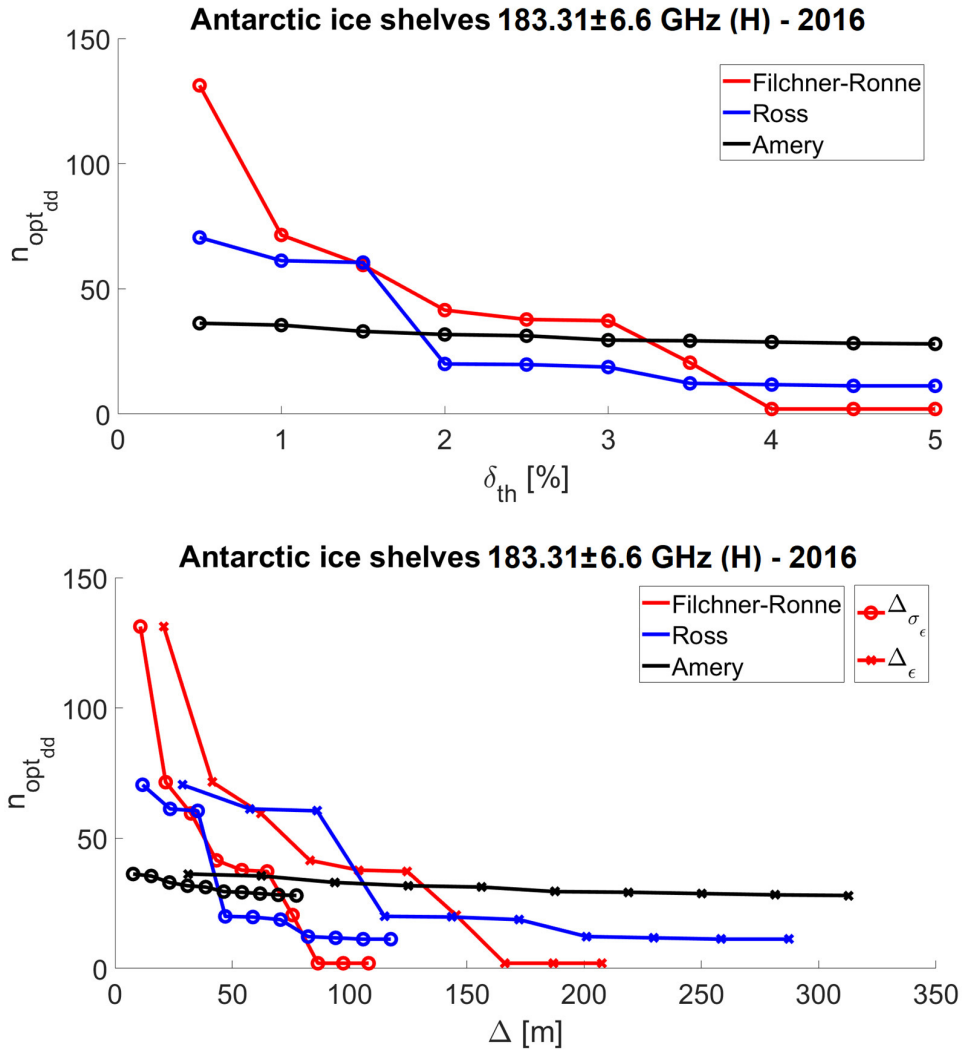


Fig. 13. (Top) Relation between the minimum optimal number $n_{\text{opt}_{\text{dd}}}$ of necessary days and δ_T threshold. (Bottom) Relation between minimum optimal number $n_{\text{opt}_{\text{dd}}}$ of necessary days and Δ_ε with "x" markers and between minimum optimal number $n_{\text{opt}_{\text{dd}}}$ of necessary days and $\Delta_{\sigma_\varepsilon}$ with "o" markers. The results are shown for the Ross ice shelf with red line, for the Filchner–Ronne ice shelf with blue line, and the Amery ice shelf with black line.

simultaneously that

$$\delta_\varepsilon(n) < \delta_{\varepsilon\text{th}} \tag{7a}$$

$$\delta_\sigma(n) < \delta_{\sigma\text{th}} \tag{7b}$$

where $\delta_{\varepsilon\text{th}}$ is the threshold for the average of geolocation displacement, $\delta_{\sigma\text{th}}$ is the threshold for the average of standard deviation of geolocation displacement. Considering for simplicity the same percentage threshold δ_{th} for (7a) and (7b),

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TABLE VI

SUMMARY OF THE GEOLOCATION ERROR ACCURACY RESULTS IN TERMS OF THE AVERAGE AND STANDARD DEVIATION FOR ALL SELECTED TARGETS DURING 2016 USING SSMIS F17 AT 183.31 ± 6.6 GHz AT HORIZONTAL POLARIZATION. SAMPLE YEARLY NUMBER AND NOTES ABOUT TARGET FEATURES ARE ALSO REPORTED

Target	Geolocation accuracy mean value [km]	Geolocation accuracy standard deviation [km]	Cloud-masked yearly sample number (percentage)	Notes
High altitude lakes and high-latitude sea bay				
Qinghai Lake	5.86	2.59	84 (13.4%)	All shift directions are sampled due to the close contour.
Lake Titicaca	6.53	2.82	41 (7.7%)	All shift directions are sampled due to the close contour.
Hudson Bay	6.29	2.89	62 (13.6%)	All shift directions are sampled due to the U contour.
average value	*6.2 km*	*2.8 km*		
Antarctic ice shelves				
Ross ice shelf	4.17	2.16	599 (25.7%)	Sharp high-resolution contour, but mainly horizontal pattern.
Filchner-Ronne ice shelf	5.75	2.35	387 (16.4%)	Sharp high-resolution contour with a V contour
Amery ice shelf	6.25	1.55	153 (12.3 %)	Sharp high-resolution contour with a nearly-vertical contour
average value	*5.5 km*	*2.0 km*		

it is possible to write the following overall condition for the minimum optimal number n_{opt}:

$$n_{\text{opt}} = n|[\delta_{\varepsilon}(n) < \delta_{\text{th}} \quad \delta_{\sigma}(n) < \delta_{\text{th}}]. \tag{8}$$

The number n_{opt} of samples that satisfies the previous (8) could be considered as the minimum optimal number of target necessary samples to reach a convergence for both the mean and standard deviation of the geolocation error. Moreover, considering the detectability/day number n_{dd} for each target, it is possible to transform n_{opt} into the minimum optimal number n_{optdd} of necessary days, needed to satisfy (8), in the following way:

$$n_{\text{opt}_{\text{dd}}} = \frac{n_{\text{opt}}}{n_{\text{dd}}}. \tag{9}$$

For example, in the case of the two lakes and the Hudson Bay, whose detectability/day is about 1, we obtain $n_{\text{opt}_{\text{dd}}}\text{dd} = n_{\text{opt}}$.

Fig. 11 shows the relation between $n_{\text{opt}_{\text{dd}}}$ and δ_{th} where δ_{th} assumes values from 0.5% to 5% with a step of 0.5%. As expected, relaxing the threshold, n_{opt} decreases. Since δ_{th} is a percentage value, then the finite difference (in km) between $\varepsilon(n_{\text{opt}})$ and m_{ε} and between $\sigma_{\varepsilon}(n)$ and m_{σ} samples depends on the target, as shown in the following equations:

$$\Delta\varepsilon = |\varepsilon(n_{\text{opt}}) - m_{\varepsilon}| = \left|\frac{\delta_{\text{th}}}{100} m_{\varepsilon}\right| \tag{10a}$$

$$\Delta\sigma_{\varepsilon} = |\sigma_{\varepsilon}(n_{\text{opt}}) - m_{\sigma}| = \left|\frac{\delta_{\text{th}}}{100} m_{\sigma}\right|. \tag{10b}$$

Fig. 11 also reports the difference expressed in (10a) and (10b). Note that $\Delta\sigma_{\varepsilon}$ is lower with respect to $\Delta\varepsilon$ for each δ_{th}, because $m_{\sigma} < m_{\varepsilon}$.

A similar approach can be carried out for all other considered targets. Fig. 12 reports the results and the minimum number of necessary samples for the Antarctic ice shelves. Considering (9) with $n_{\text{dd}} = 4$ for ice shelves, it is possible to obtain Fig. 13, showing the minimum number of days necessary to obtain the convergence. Note that the $n_{\text{opt}_{\text{dd}}}$ becomes similar, for the three shelves, for $\delta_{\text{th}} \geq 2\%$. For example, for ice shelves with $\delta_{\text{th}} = 2\%$, almost 40 days are needed for the convergence of the geolocation error moments.

V. CONCLUSION

The goal of this study is to propose a systematic methodology to validate the geolocation error for channel at 183.31 GHz

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TABLE VII
COORDINATES OF THE FIVE POINTS USED TO CALCULATE THE BT CONTRAST FOR THE LAKE TITICACA AND QINGHAI LAKES, FOLLOWING (11)

Pixels	Lake Titicaca		Qinghai Lake	
	Latitude [deg]	**Longitude** [deg]	**Latitude** [deg]	**Longitude** [deg]
A	-15.8766	-69.3462	36.9500	100.1793
B	-15.1745	-69.3462	37.3719	100.1793
C	-15.8766	-68.6250	36.9500	100.7655
D	-16.5787	-69.3462	36.5750	100.1793
E	-15.8766	-70.2115	36.9500	99.5345

TABLE VIII
COORDINATES OF THE EIGHT POINTS USED TO CALCULATE THE BT CONTRAST FOR THE HUDSON BAY, FOLLOWING (12)

Pixels	Hudson Bay	
	Latitude [deg]	**Longitude** [deg]
A	61.4091	-94.8545
B	61.4091	-92.8091
C	54.1364	-95.4273
D	54.1364	-93.7909
E	57.3182	-92.9727
F	57.7727	-91.9909
G	56.4091	-89.1273
H	57.0909	-88.7182

using the 183.31 $\pm$ 6.6 GHz (H), including the criteria regarding the selection of landmark targets and the automated cloud-masking defuzzification step to filter the available data set from cloud coverage contamination. The geolocation making use of this sounding channel is assessed in view of the launch of the EPS-SG ICI instrument, whose lower frequency is at 183.31 GHz. As such, the geolocation performance of this instrument cannot be inferred from the straightforward application of the techniques developed so far for microwave imager radiometers using lower frequency high-surface-sensitive channels. A total number of six landmark targets (four in the southern hemisphere and two in the northern one) have been discussed in this study to prove the methodology and highlight the peculiarities of the targets and atmospheric/seasonal conditions beneath their applicability to the geolocation assessment.

Table VI reports a summary of the geolocation accuracy validation for all targets using SSMIS F17 at 183.31 $\pm$ 6.6 GHz at horizontal polarization. The average values are about 6.2 and 5.4 km for mountainous lakes and Hudson Bay and ice shelves, respectively, with a standard deviation of about 2.7 and 2.0 km. These numbers can be interpreted as the mean geolocation error of SSMIS-selected imagery. These results are comparable with the values given in [8] and [31] for SSMIS F16 (different from F17 we have used in this work) even though they mention the estimate less than 6 km [31]. Note that Poe *et al.* [10] referred to their estimate as 1-sigma error value, whereas Kunkee *et al.* [31] stressed the fairly good stability of their retrieved error.

Additional purpose of this work is to demonstrate that SAR images represent a promising source of reference for geolocation algorithms. SAR images allowed the use of slowly varying targets as ice shelves. Indeed, the results, discussed in Section IV, show comparable geolocation accuracy for three Antarctic ice shelves and for two high-altitude lakes, in which we used the reference GSHHG shoreline database.

APPENDIX
FUZZY-LOGIC APPROACH TO CLOUD MASKING

In the proposed fuzzy-logic approach [32], the idea is to use the estimated geolocation error and BT contrast of a specific target to understand if an overpass can be correctly used. For this purpose, we define the membership function $M_1(\varepsilon)$, which is the membership function depending on the geolocation

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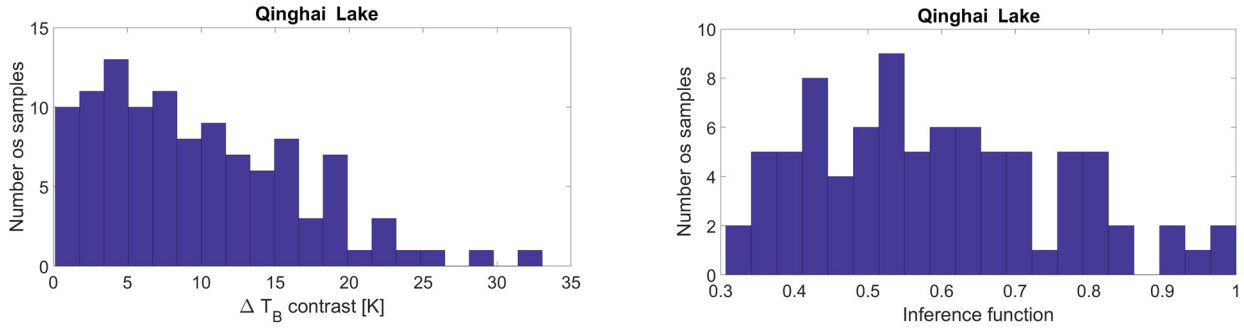


Fig. 14. (Left) Histogram of ΔT_{Bm} values for the Qinghai Lake, for the data set without samples with negative ΔT_{Bm} and with $M_1(\varepsilon) = 0$. (Right) Histogram of inference function for the Qinghai Lake with the final data set of 84 images.

TABLE IX
POLAR STEREOGRAPHIC COORDINATES OF THE EIGHT POINTS USED TO CALCULATE THE BT CONTRAST FOR THREE ICE SHELVES, FOLLOWING (12)

Pixels	Ross		Filchner-Ronne		Amery	
	X	Y	X	Y	X	Y
A	6.3796 10^4	-1.3149 10^6	-1.5011 10^6	8.3170 10^5	2.1984 10^6	7.6657 10^5
B	6.3796 10^4	-1.3763 10^6	-1.5011 10^6	8.8246 10^5	2.2356 10^6	7.6657 10^5
C	1.0524 10^5	-1.33 10^6	-1.4453 10^6	7.9110 10^5	2.2303 10^6	7.1870 10^5
D	1.0524 10^5	-1.4070 10^6	-1.4453 10^6	8.2663 10^5	2.2675 10^6	7.1875 10^5
E	1.6222 10^5	-1.3303 10^6	-1.3794 10^6	8.0633 10^5	2.2515 10^6	6.7093 10^5
F	1.6222 10^5	-1.3814 10^6	-1.3794 10^6	8.4693 10^5	2.2143 10^6	6.7093 10^5
G	2.1920 10^5	-1.3251 10^6	-1.3185 10^6	8.4186 10^5	2.1984 10^6	6.3373 10^5
H	2.1920 10^5	-1.3712 10^6	-1.3185 10^6	8.7739 10^5	2.2356 10^6	6.3373 10^5

error (ε), and $M_2(\Delta T_{\mathrm{Bm}})$, which depends on the mean BT contrast around target ΔT_{Bm}. The corresponding equation for both functions is described as follows:

$$M_1(\varepsilon) = \begin{cases} \text{linear}, & \text{if } \varepsilon \le \varepsilon_{\mathrm{th}} \\ 0, & \text{if } \varepsilon > \varepsilon_{\mathrm{th}} \end{cases} \tag{A1}$$

where $\varepsilon_{\mathrm{th}}$ is the geolocation error threshold and $\Delta T_{Bm_{\mathrm{th}}}$ is the mean BT contrast around target threshold

$$M_2(\Delta T_{\mathrm{Bm}}) = \begin{cases} \text{linear}, & \text{if } \Delta T_{\mathrm{Bm}} \le \Delta T_{\mathrm{Bm_{th}}} \\ 1, & \text{if } \Delta T_{\mathrm{Bm}} > \Delta T_{\mathrm{Bm_{th}}}. \end{cases} \tag{A2}$$

If the geolocation error is smaller than a selected threshold or the BT contrast is lower than a specific threshold, the membership functions are linearly weighted. The proposed inference function is

$$I(\varepsilon, \Delta T_{\mathrm{Bm}}) = M_1(\varepsilon) M_2(\Delta T_{\mathrm{Bm}}). \tag{A3}$$

Finally, an image can be used to evaluate the geolocation error if it satisfies the following defuzzification step:

$$I(\varepsilon, \Delta T_{\mathrm{Bm}}) \ge I_{\mathrm{threshold}} \tag{A4}$$

where $I_{\mathrm{threshold}}$ is typically set to 0.3.

The parameter ε is evaluated using the proposed TCM algorithm and ΔT_{Bm} is obtained differently for each target. For high-altitude lakes, the ΔT_{Bm} is calculated by (10)

$$\Delta T_{\mathrm{Bm}} = \frac{(T_{\mathrm{B}} - T_{\mathrm{A}}) + (T_{\mathrm{C}} - T_{\mathrm{A}}) + (T_{\mathrm{D}} - T_{\mathrm{A}}) + (T_{\mathrm{E}} - T_{\mathrm{A}})}{4} \tag{A5}$$

where T_{A}, T_{B}, T_{C}, T_{D} and T_{E} are the BT of the five points indicated in Table VII for the Lake Titicaca and Qinghai Lake.

The BT contrast for ice shelves and the Hudson Bay has been calculated through (11) with points in Tables VIII and IX, expressed on polar stereographic map

$$\Delta T_{\mathrm{Bm}} = \frac{(T_{\mathrm{B}} - T_{\mathrm{A}}) + (T_{\mathrm{D}} - T_{\mathrm{C}}) + (T_{\mathrm{F}} - T_{\mathrm{E}}) + (T_{\mathrm{H}} - T_{\mathrm{G}})}{4}. \tag{A6}$$

To establish the threshold for $M_1(\varepsilon)$, we considered the pixel size for simplification, which is about 13 $\times$ 16 km^2 in case of SSMIS F17 187 $\pm$ 6.6 GHz. We can then expect a geolocation accuracy lower than this value and we set 15 km as a threshold for $M_1(\varepsilon)$ for all targets. To establish the threshold for $M_2(\Delta T_{\mathrm{Bm}})$, we have observed the BT contrast of various targets.

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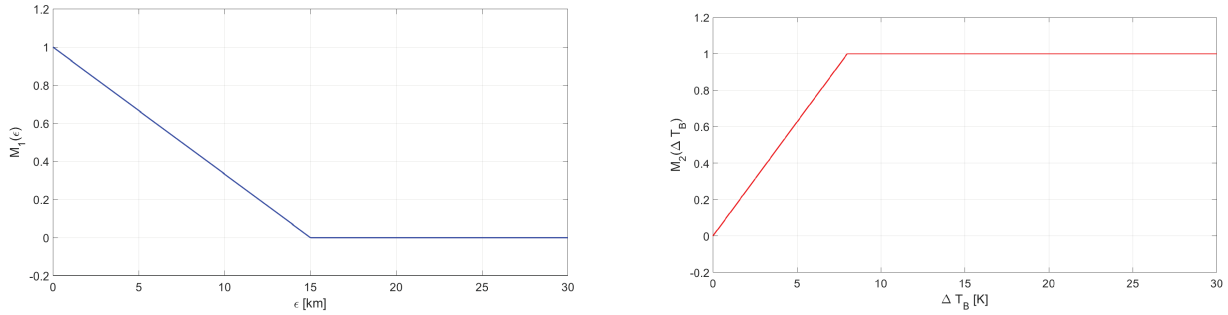


Fig. 15. Blue line (Left) represent the $M_1(\varepsilon)$ function and the red line (Right) represent the $M_2(\Delta T_{\mathrm{Bm}})$ function for the Qinghai Lake.

As an example, the Qinghai Lake has a total of 624 overpasses along 2016. To obtain the threshold, initially we neglected the images with negative ΔT_{Bm}, calculated following (10), and then we removed the samples with $M_1(\varepsilon) = 0$. The resulting data set has the ΔT_{Bm} values shown in Fig. 14 (left). The median value of that histogram is about 8.4 K, so we have chosen a value of 8 K as threshold for $M_2(\Delta T_{\mathrm{Bm}})$. Fig. 15 reports the behavior of the function $M_1(\varepsilon)$ and $M_2(\Delta T_{\mathrm{Bm}})$ used for the Qinghai Lake. For $M_1(\varepsilon)$, the linear function ranges from 0 to 15 km, while $M_1(\varepsilon)$ is equal to 0 for $\varepsilon > 15$ km. $M_2(\Delta T_{\mathrm{Bm}})$ has linear behavior from 0 to 8 K and then, for $\Delta T_{\mathrm{Bm}} > 8$ K, it is equal to 1. After the defuzzification step in (9), the total data set is represented by 84 samples, obtaining the inference function of (8) shown in Fig. 14 (right). The value of $I(\varepsilon, \Delta T_{\mathrm{Bm}})$ determines the goodness of an image so that the SSMIS images with this value close to 1 represent the optimal samples.

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